

DevOps Engineering

DevOps Engineering: Reference-1

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239	Server Usage – Topics 2
240	Cloud Storage Device Usage – Topics 2
241	Cloud Service Usage – Topics 3
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243	Pricing Models
244	Additional Considerations Case Study Example
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250	Service Availability Metrics – Topics 2
251	Service Reliability Metrics – Topics 2
252	Service Performance Metrics – Topics 7
253	Service Scalability Metrics – Topics 3
254	Service Resiliency Metrics – Topics 2
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257	Case Study Example – Topics 5
	Virtualization Essential

DevOpa (Tools)

Learning DevOps can feel overwhelming, but with a structured plan, it becomes manageable. Below is a comprehensive, step-by-step **6-month guide** that covers everything from project management to monitoring, including all the tools we discussed.

This schedule is designed for a commitment of **15–20 hours per week**. Adjust the pace to fit your availability—it can be stretched to 8–9 months if you have less time.

 The 6-Month DevOps Learning Schedule

Phase	Month	Week	Topic	Key Tools & Technologies	Hands-On Project / Task
1. Foundations	1	1	DevOps Culture & Project Management	Agile, Scrum, Kanban, Jira, Trello, Confluence	Set up a Jira project board. Create user stories and a sprint backlog for a sample application.
		2	Linux Fundamentals & Networking	Linux CLI (Ubuntu/CentOS), Bash, SSH, Permissions, Systemd	Navigate the file system, manage users, set up a web server (Nginx/Apache) manually.
		3	Version Control with Git	Git, GitHub, GitLab, Branching (Git Flow)	Create a repo, commit changes, create branches, merge, and handle merge conflicts.
		4	Scripting & Automation	Bash Scripting, Python Basics	Write a Bash script to automate system backups. Write a Python script to parse log files.
Phase	Month	Week	Topic	Key Tools & Technologies	Hands-On Project / Task
2. CI/CD & Build	2	5	Build Systems & Package Management	C++: CMake, Conan Python: Poetry, pip JavaScript: Vite, npm	Set up a build system for a sample project in your chosen language. Manage dependencies.
		6	Testing Fundamentals	C++: Google Test Python: pytest JavaScript: Jest	Write unit tests and integration tests for your sample project. Aim for >80% code coverage.
		7	Continuous Integration (CI)	Jenkins, GitHub Actions, GitLab CI	Create a CI pipeline that automatically builds and tests your code on every push.
		8	Continuous Delivery (CD)	GitHub Actions, GitLab CI, ArgoCD, Spinnaker	Extend your pipeline to automatically deploy your application to a staging environment.
Phase	Month	Week	Topic	Key Tools & Technologies	Hands-On Project / Task

3. Containers & Orchestration	3	9	Containerization with Docker	Docker, Docker Compose, Container Registries	Containerize your sample application. Use Docker Compose to run multi-container apps.
		10	Container Orchestration (Intro)	Kubernetes (K8s), Minikube, kubectl	Deploy a containerized app to a local Kubernetes cluster. Learn Pods, Deployments, Services.
		11	Advanced Kubernetes	Helm, Ingress, Persistent Volumes, ConfigMaps	Package your app with Helm. Set up ingress for external access. Manage configuration.
		12	Kubernetes in Production	K3s, EKS, AKS, GKE	Deploy a scalable, highly available application on a managed Kubernetes service (e.g., EKS).
Phase	Month	Week	Topic	Key Tools & Technologies	Hands-On Project / Task
4. Infrastructure as Code	4	13	Configuration Management	Ansible, Chef, Puppet	Use Ansible to automate the setup of a web server (install Nginx, configure firewall).
		14	Infrastructure as Code (IaC)	Terraform, Pulumi, AWS CloudFormation	Provision a cloud environment (VPC, subnets, EC2 instances) using Terraform.
		15	Cloud Platforms (AWS/Azure/GCP)	AWS (EC2, S3, RDS, VPC), Azure, GCP	Deploy a sample application to the cloud. Use managed services like RDS for a database.
		16	GitOps	ArgoCD, Flux	Implement GitOps: use Git as the single source of truth for your infrastructure and application state.
Phase	Month	Week	Topic	Key Tools & Technologies	Hands-On Project / Task
5. Monitoring & Debugging	5	17	Logging	ELK Stack (Elasticsearch, Logstash, Kibana), Loki	Centralize logs from your application and infrastructure. Create dashboards in Kibana.
		18	Metrics & Monitoring	Prometheus, Grafana	Set up Prometheus to collect metrics (CPU, memory, request latency). Visualize them in Grafana.
		19	Alerting & Incident Management	Alertmanager, PagerDuty, Opsgenie	Configure alerts for high error rates or system downtime. Integrate with PagerDuty for on-call rotations.
		20	Debugging & Distributed Tracing	Jaeger, Zipkin, Sentry, GDB/Valgrind (C++)	Implement distributed tracing to track requests

					across microservices. Use Sentry for error tracking.
Phase	Month	Week	Topic	Key Tools & Technologies	Hands-On Project / Task
6. Advanced & Integration	6	21	Security (DevSecOps)	SonarQube, Snyk, Trivy, OWASP ZAP	Integrate security scanning into your CI/CD pipeline (SAST, DAST, dependency scanning).
		22	AI & MLOps (Optional)	MLflow, Kubeflow, NVIDIA Triton, DVC	Learn how DevOps principles apply to machine learning. Set up a model training and serving pipeline.
		23	Final Project (Part 1)	All tools combined	Project: Design and implement a complete end-to-end CI/CD pipeline for a microservices application on Kubernetes with monitoring and alerting.
		24	Final Project (Part 2)	All tools combined	Project: Polish the project, document everything, and present it as a portfolio piece.

The Recommended Learning Path

This schedule follows a proven, logical progression:

1. **Start with the Fundamentals:** Linux, networking, and a scripting language like **Bash** or **Python** are your foundation. Without these, the rest will be very difficult.
2. **Master Version Control:** Get comfortable with **Git** and a platform like **GitHub** or **GitLab**. This is non-negotiable.
3. **Learn Build & Test:** Understand how to compile code and run tests automatically. This prepares you for CI/CD.
4. **Build CI/CD Pipelines:** Use a tool like **GitHub Actions** or **Jenkins** to automate the build, test, and deployment process.
5. **Learn Containerization:** Begin with **Docker** to understand how to package applications.
6. **Move to Orchestration:** Once comfortable with Docker, learn **Kubernetes** to manage containers at scale.
7. **Tackle Infrastructure as Code:** Learn **Terraform** to provision the infrastructure your containers will run on.
8. **Implement Monitoring & Logging:** Finally, learn **Prometheus** and **Grafana** to observe your running applications.

💡 Tips for Success

- **Hands-on > Theory:** DevOps is a practical field. Spend at least 70% of your time doing hands-on labs and projects.
- **One Tool at a Time:** Don't try to learn everything at once. Master one tool before moving to the next.
- **Use Free Resources:**
 - **Roadmap.sh:** The [DevOps Roadmap](#) is an excellent visual guide.
 - **GitHub:** Explore the many free DevOps learning paths and resources available.
 - **Cloud Free Tiers:** AWS, Azure, and GCP all offer free tiers perfect for learning.
- **Build a Portfolio:** The final project is crucial. It demonstrates your ability to apply all the tools in a real-world scenario.
- **Join a Community:** Engage with DevOps communities on Reddit (r/devops), LinkedIn, or Discord. Ask questions and learn from others.

🤔 What About My Programming Language?

You can apply this roadmap with **any** language (C++, Python, JavaScript, etc.). The tools for **build, test, and package management** will change, but the CI/CD, containerization, and cloud concepts remain the same. Choose one language to focus on for your projects.

This schedule provides a clear, step-by-step path. Would you like me to elaborate on any specific week or tool?

Below is a comprehensive, end-to-end DevOps tools table, classified from **Project Management** to **Monitoring**. For each category, I've highlighted the **most popular** tools specifically for **C++**, **Python**, **Web (JavaScript)**, and **AI/ML** development in 2025.

 The Complete DevOps Toolchain (Classified by Stage)

Stage	Category	C++	Python	JS/TypeScript	AI / ML
Plan	Project Management	Jira, Linear, Azure DevOps	Jira, Linear, Azure DevOps	Jira, Linear, Azure DevOps	Jira, Linear, Azure DevOps
Plan	Documentation & Knowledge	Confluence, Notion, GitHub Wiki	Confluence, Notion, GitHub Wiki	Confluence, Notion, GitHub Wiki	Confluence, Notion, GitHub Wiki
Code	Version Control (SCM)	Git, GitHub, GitLab, Bitbucket	Git, GitHub, GitLab, Bitbucket	Git, GitHub, GitLab, Bitbucket	Git, GitHub, GitLab, Bitbucket
Code	IDE & Local Environment	Visual Studio, CLion, VS Code	PyCharm, VS Code, Jupyter	VS Code, WebStorm, Cursor	VS Code, Jupyter, Cursor
Code	AI Coding Assistant	GitHub Copilot, SourceCraft	GitHub Copilot, Amazon Q	GitHub Copilot, Cursor AI	GitHub Copilot, Cursor AI, Sourcegraph Cody
Build	Package Manager	Conan, vcpkg, Build2	pip, Poetry, Conda	npm, yarn, pnpm	pip, Conda, Poetry
Build	Build System / Compiler	CMake, Ninja, Bazel, Make	Setuptools, Poetry, PyInstaller	Vite, Webpack, esbuild, Rollup	Setuptools, Poetry, Bazel
Build	Code Quality & Linting	Clang-Tidy, Clang-Format, SonarQube	Ruff, Flake8, Black, mypy	ESLint, Prettier, SonarQube	Ruff, Pylint, mypy
Test	Unit / Integration Testing	Google Test, Catch2, CTest	pytest, unittest, nose	Jest, Vitest, Mocha	pytest, unittest
Test	E2E / Browser Automation	Selenium, Playwright	Selenium, Playwright	Playwright, Cypress, Selenium	Selenium, Playwright
Test	Performance / Load Testing	JMeter, Gatling, Valgrind	JMeter, Locust	JMeter, k6	JMeter, k6
Test	Security / SAST	SonarQube, CodeQL	Bandit, SonarQube, CodeQL	SonarQube, CodeQL, Snyc	Bandit, SonarQube, CodeQL
CI	Continuous Integration	GitHub Actions, GitLab CI, Jenkins	GitHub Actions, GitLab CI, Jenkins	GitHub Actions, GitLab CI, CircleCI	GitHub Actions, GitLab CI, Jenkins
CD	Continuous Delivery / GitOps	ArgoCD, Flux	ArgoCD, Flux, Spinnaker	ArgoCD, Flux, Vercel, Netlify	ArgoCD, Flux, Kubeflow
Deploy	Containerization	Docker, Podman	Docker, Podman	Docker, Podman	Docker, Podman, NVIDIA Container Toolkit
Deploy	Container Orchestration	Kubernetes, K3s	Kubernetes, K3s	Kubernetes, K3s	Kubernetes, K3s

Deploy	Infrastructure as Code (IaC)	Terraform, Pulumi	Terraform, Pulumi, AWS CDK	Terraform, Pulumi	Terraform, Pulumi
Deploy	Configuration Management	Ansible, Chef	Ansible, SaltStack, Fabric	Ansible	Ansible
Deploy	ML Model Serving	—	—	—	NVIDIA Triton, TensorFlow Serving, BentoML
Deploy	ML Pipeline Orchestration	—	—	—	Apache Airflow, Kubeflow, MLflow, Metaflow
Operate	Debugging & Tracing	GDB, Valgrind, AddressSanitizer	pdb, py-spy, Sentry	Chrome DevTools, Sentry	LangSmith, Weights & Biases
Operate	Distributed Tracing	Jaeger, Zipkin	Jaeger, Zipkin	Jaeger, Zipkin	Jaeger, Zipkin
Monitor	Metrics & Monitoring	Prometheus + Grafana	Prometheus + Grafana	Prometheus + Grafana	Prometheus + Grafana
Monitor	Logging	ELK Stack (Elasticsearch, Logstash, Kibana)	ELK Stack	ELK Stack	ELK Stack
Monitor	Observability Platforms	Datadog, New Relic, Dynatrace	Datadog, New Relic, Dynatrace	Datadog, New Relic, Dynatrace	Datadog, New Relic, Dynatrace
Monitor	Incident Alerting	Alertmanager, PagerDuty, Opsgenie	Alertmanager, PagerDuty, Opsgenie	Alertmanager, PagerDuty, Opsgenie	Alertmanager, PagerDuty, Opsgenie

AI's Role IN DevOps (Not Just FOR AI)

AI is now embedded **inside** DevOps tools themselves:

Tool	AI Feature
GitHub Copilot	Writes YAML pipelines, Dockerfiles, and Terraform code.
Amazon Q Developer	Expert in IaC (Terraform/CDK) and AWS.
Dynatrace / Datadog	AI-powered anomaly detection and root cause analysis.
Snyk AI	AI-assisted security vulnerability fixing.

🔑 Key Observations by Language

Language	Build System	Testing	CI/CD	Special Notes
C++	CMake + Conan/vcpkg is the gold standard. Bazel for massive monorepos.	Google Test for unit tests. Valgrind & AddressSanitizer for memory debugging.	GitHub Actions & GitLab CI are most popular.	C++ DevOps is heavily focused on build speed (Ninja, ccache) and dependency management .
Python	Poetry is modern standard. Setuptools for legacy.	pytest is the undisputed king. Ruff is the new ultra-fast linter.	GitHub Actions is #1. Ansible & Fabric for automation.	Python shines in automation scripts and cloud SDKs (boto3, kubernetes client).
JavaScript	Vite is now #1 (98% usage). esbuild (88%) and Rollup (79%) are also huge. Webpack is declining (55%).	Jest is most popular. Vitest is gaining fast.	GitHub Actions & GitLab CI dominate. Vercel/Netlify for frontend hosting.	JavaScript DevOps is about fast builds and frontend deployment . TypeScript is now assumed.
AI / ML	Conda & Poetry for Python dependencies.	pytest for code. Great Expectations for data validation.	GitHub Actions for code CI. Kubeflow, Airflow, MLflow for ML pipelines.	MLOps is a superset of DevOps. Key additions: Model Serving (Triton, TF Serving), Experiment Tracking (MLflow, Weights & Biases), and Data Versioning (DVC, LakeFS).

📌 Summary

Your Original List	DevOps Category	Tools (2025)
Git	Version Control	Git, GitHub, GitLab
C++, Python, JavaScript	Build + Test + Code Quality	CMake/Conan (C++), Poetry/pytest (Python), Vite/Jest (JS)
Web-based environment, IDE	Local Dev Environment	VS Code, Cursor, WebStorm, PyCharm
Library, packages setup	Package Managers	Conan/vcpkg, pip/Poetry, npm/yarn
Build system	Build Tools	CMake/Ninja, Vite/esbuild, Setuptools
Integration	CI/CD	GitHub Actions, GitLab CI, Jenkins
Cloud computing	IaC + Cloud	Terraform, Kubernetes, AWS/Azure/GCP
Testing	Testing Frameworks	Google Test, pytest, Jest
Debugging	Debugging + Tracing	GDB/Valgrind, pdb/Sentry, Jaeger/Zipkin
Design Management	Architecture (Not DevOps)	Handled by Software Architects
Consumer Satisfaction	Product Management (Not DevOps)	Handled by PMs & UX Researchers
Project Management	Planning & Tracking	Jira, Linear, Azure DevOps
Monitoring	Observability	Prometheus + Grafana, ELK, Datadog

This table gives you a **complete, language-specific roadmap** for mastering DevOps in 2025. Would you like me to go deeper into any specific category (e.g., how to set up a CI/CD pipeline for C++ with CMake + GitHub Actions)?

Calculus and Analytical Geometry

Calculus and Analytical Geometry: Reference-1

	Calculus and Analytic Geometry
	Preliminaries
07	Real Numbers and the Real Line – Topics 7
15	Coordinates, Lines and Increments – Topics 8
24	Functions – Topics 9
29	Shifting Graphs – Topics 5
34	Trigonometric Functions – Topics 5
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	Additional Exercises – Theory, Examples, Applications
	Limits and Continuity
37	Rates of Change and Limits – Topics 3
41	Rules for Finding Limits – Topics 4
47	Target Values and Formal Definitions of Limits – Topics 6
52	Extensions of the Limit Concept – Topics 5
60	Continuity – Topics 8
63	Tangent Lines – Topics 3
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	Additional Exercises – Theory, Examples, Applications
	Derivatives
70	The Derivative of a Function – Topics 7
78	Differentiation Rules – Topics 8
83	Rates of Change – Topics 5
93	Derivatives of Trigonometric Functions – Topics 10
99	The Chain Rule – Topics 6
103	Implicit Differentiation and Rational Exponents – Topics 4
105	Related Rates of Change – Topics 2
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	Additional Exercises – Theory, Examples, Applications
	Applications of Derivatives
111	Extreme Values of Functions – Topics 6
119	The Mean Value Theorem – Topics 8
120	The First Derivative Test for Local Extreme Values – Topics 1
125	Graphing with y' and y'' – Topics 5
134	Limits as $x \rightarrow +\infty$, Asymptotes, and Dominant Terms – Topics 9
141	Optimization – Topics 7
149	Linearization and Differentials – Topics 8
154	Newton's Method – Topics 5
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	Integration
158	Indefinite Integrals – Topics 4
162	Differential Equations, Initial Value Problems, and Mathematical Modeling – Topics 4
165	Integration by Substitution – Running the Chain Rule Backward – Topics 3
170	Estimating with Finite Sums – Topics 5
180	Riemann Sums and Definite Integrals – Topics 10
184	Properties, Area, and the Mean Value Theorem – Topics 4
188	The Fundamental Theorem – Topics 4

189	Substitution in Definite Integrals – Topics 1
196	Numerical Integration – Topics 7
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202	Areas Between Curves – Topics 6
204	Finding Volumes by Slicing – Topics 2
208	Volumes of Solids of Revolution – Disks and Washers – Topics 4
210	Cylindrical Shells – Topics 2
214	Lengths of Plane Curves – Topics 4
218	Areas of Surfaces of Revolution – Topics 4
222	Moments and Centers of Mass – Topics 4
226	Work – Topics 4
232	Fluid Pressures and Forces – Topics 6
237	The Basic Pattern and Other Modeling Applications – Topics 5
	Questions to Guide Your Review and Practice Exercise
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	Transcendental Functions
243	Inverse Functions and Their Derivatives – Topics 6
250	Natural Logarithms – Topics 7
254	The Exponential Function – Topics 4
263	Ax and $\log_a X$ – Topics 9
267	Growth and Decay – Topics 4
272	L'Hopital's Rules – Topics 5
276	Relative Rates of Growth – Topics 4
279	Inverse trigonometric Functions – Topics 3
283	Derivatives of Inverse trigonometric Functions; Integrals – Topics 4
287	Hyperbolic Functions – Topics 4
291	First Order Differential Equations – Topics 4
265	Euler's Numerical Method: Slope Fields – Topics 4
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	Additional Exercises – Theory, Examples, Applications
	Techniques of Integration
267	Basic Integration Formulas – Topics 2
271	Integration by Parts – Topics 4
275	Partial Fractions – Topics 4
276	Trigonometric Substitutions – Topics 1
280	Integral tables and CAS – Topics 4
281	Improper Integrals – Topics 1
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	Additional Exercises – Theory, Examples, Applications
	Infinite Series
285	Limits of Sequences of Numbers – Topics 4
293	Theorems for Calculation Limits of Sequences – Topics 7
298	Infinite Series – Topics 5
30	The Integral Test for Series of Nonnegative Terms – Topics 2
302	Comparison Tests for Series of Nonnegative Terms – Topics 2
304	The Ratio and Root Tests for Series of Nonnegative Terms – Topics 2
306	The Ratio and Root Tests for Series of Nonnegative Terms – Topics 2
310	Alternating Series, Absolute and Conditional Convergence – Topics 4
315	Power Series – Topics 5

318	Taylor and Maclaurin Series – Topics 3
323	Convergence of Taylor Series; Error Estimates – Topics 5
328	Applications of Power Series – Topics 5
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	Conic Sections, Parametrized Curves, and Polar Coordinates
336	Conic Sections and Quadratic Equations – Topics 8
339	Classifying Conic Sections by Eccentricity – Topics 3
343	Quadratic Equations and Rotations – Topics 4
345	Parametrizations of Plane Curves – Topics 2
349	Calculus with Parametrized Curves – Topics 4
353	Polar Coordinates – Topics 4
358	Graphing in Polar Coordinates – Topics 5
361	Polar Equations for Conic Sections – Topics 3
364	Integration in Polar Coordinates – Topics 3
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381	Cartesian (Rectangular) Coordinates and Vectors in Space – Topics 8
386	Dot Products Cross Products – Topics 5
390	Lines and Planes in Space – Topics 4
395	Cylinders and Quadric Surfaces – Topics 5
397	Cylindrical and Spherical Coordinates – Topics 2
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405	Vector-Valued Functions and Space Curves – Topics 8
408	Modeling Projectile Motion – Topics 3
410	Arc Length and the Unit Tangent Vector – Topics 2
416	Curvature, Torsion, and the TNB Frame – Topics 6
422	Planetary Motion and Satellites – Topics 6
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433	Limits and Continuity – Topics 6
438	Partial Derivatives – Topics 5
442	Differentiability, Linearization, and Differentials – Topics 4
447	The Chain Rule – Topics 5
450	Partial Derivatives with Constrained Variables – Topics 3
459	Directional Derivatives, Gradient Vectors, and Tangent Planes – Topics 9
463	Extreme Values and Saddle Points – Topics 4
467	Lagrange Multipliers – Topics 4
470	Taylor's Formula – Topics 3
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	Multiple Integrals
478	Double Integrals – Topics 8
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485	Double Integrals in Polar Form – Topics 3

488	Triple Integrals in rectangular Coordinates – Topics 3
492	Masses and Moments in Three Dimensions – Topics 4
496	Triple Integrals in Cylindrical and Spherical Coordinates – Topics 4
498	Substitutions in Multiple Integrals – Topics 2
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506	Vector Fields, Work, Circulation, and Flux – Topics 5
512	Path Independence, Potential Functions, and Conservative Fields – Topics 6
519	Green’s Theorem in the Plane – Topics 7
526	Surface Area and Surface Integrals – Topics 6
530	Parametrized Surfaces – Topics 4
536	Stokes’s Theorem – Topics 6
543	The Divergence Theorem and a Unified Theory – Topics 7
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	Proofs of Limit Theorems in Section
	Complex Numbers
	Simpson’s One-Third Rule
	Cauchy’s Mean Value Theorem and the Stronger Form of l’Hopital’s Rule
	Limits That Arise Frequently
	The Distributive Law for Vector Cross Products
	Determinants and Craner’s Rule
	Euler’s Theorem and the Increment Theorem

Calculus and Analytical Geometry: Reference-2

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	Ordinary Differential Equations
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06	Geometric Meaning of $y = f(x, y)$. Direction Fields, Euler's Method – Topics 2
11	Separable ODEs. Modeling – Topics 5
14	Exact ODEs. Integrating Factors – Topics 3
18	Linear ODEs. Bernoulli Equations. Population Dynamics – Topics 4
19	Orthogonal Trajectories. Optional – Topics 1
21	Existence and Uniqueness of Solutions for Initial Value Problems – Topics 2
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	Second-Order Linear ODEs
25	Homogeneous Linear ODEs of Second Order – Topics 4
30	Homogeneous Linear ODEs with Constant Coefficients – Topics 5
31	Differential Operators. Optional – Topics 1
35	Modeling of Free Oscillations of a Mass-Spring System – Topics 4
36	Euler-Cauchy Equations – Topics 1
39	Existence and Uniqueness of Solutions. Wronskian – Topics 3
43	Nonhomogeneous ODEs – Topics 4
47	Modeling: Forced Oscillations. Resonance – Topics 4
50	Modeling: Electric Circuits – Topics 3
52	Solution by Variation of Parameters – Topics 2
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57	Homogeneous Linear ODEs – Topics 5
62	Homogeneous Linear ODEs with Constant Coefficients – Topics 5
66	Nonhomogeneous Linear ODEs – Topics 4
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72	For Reference: Basics of Matrices and Vectors – Topics 6
74	Systems of ODEs as Models in Engineering Applications – Topics 2
77	Basic Theory of Systems of ODEs. Wronskian – Topics 3
82	Constant-Coefficient System. Phase Plane Method – Topics 5
83	Criteria for Critical Points. Stability – Topics 1
87	Qualitative Methods for Nonlinear Systems – Topics 4
90	Nonhomogeneous Linear Systems of ODEs – Topics 3
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96	Legendre's Equations. Legendre Polynomials $P_n(x)$ – Topics 2
99	Extended Power Series Methods: Frobenius Method – Topics 3
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107	Bessel Functions of the $Y_\nu(x)$. General Solution – Topics 2
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114	Transforms of Derivatives and Integrals. ODEs – Topics 3
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121	Short Impulses. Dirac's Delta Function. Partial Fractions – Topics 4
125	Convolution. Integral Equations – Topics 4
128	Differentiation and Integration of Transforms ODEs with Variable Coefficients – Topics 3
131	Systems of ODEs – Topics 3
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136	Matrices, Vectors: Addition and Scalar Multiplication – Topics 3
141	Matrix Multiplication – Topics 5
146	Linear Systems of Equations. Gauss Elimination – Topics 5
149	Linear Independence. Rank of a Matrix. Vector Space – Topics 3
152	Solutions of Linear Systems: Existence, Uniqueness – Topics 3
154	For Reference: Second- and Third-Order Determinants – Topics 2
157	Determinants. Cramer's Rule – Topics 3
162	Inverse of a Matrix. Gauss-Jordan Elimination – Topics 5
166	Vector Spaces, Inner Product Spaces. Linear Transformations. Optional – Topics 4
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171	Some Applications of Eigenvalue Problems – Topics 2
173	Symmetric, Skew-Symmetric, and Orthogonal Matrices – Topics 2
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179	Complex Matrices and Forms. Optional – Topics 3
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